

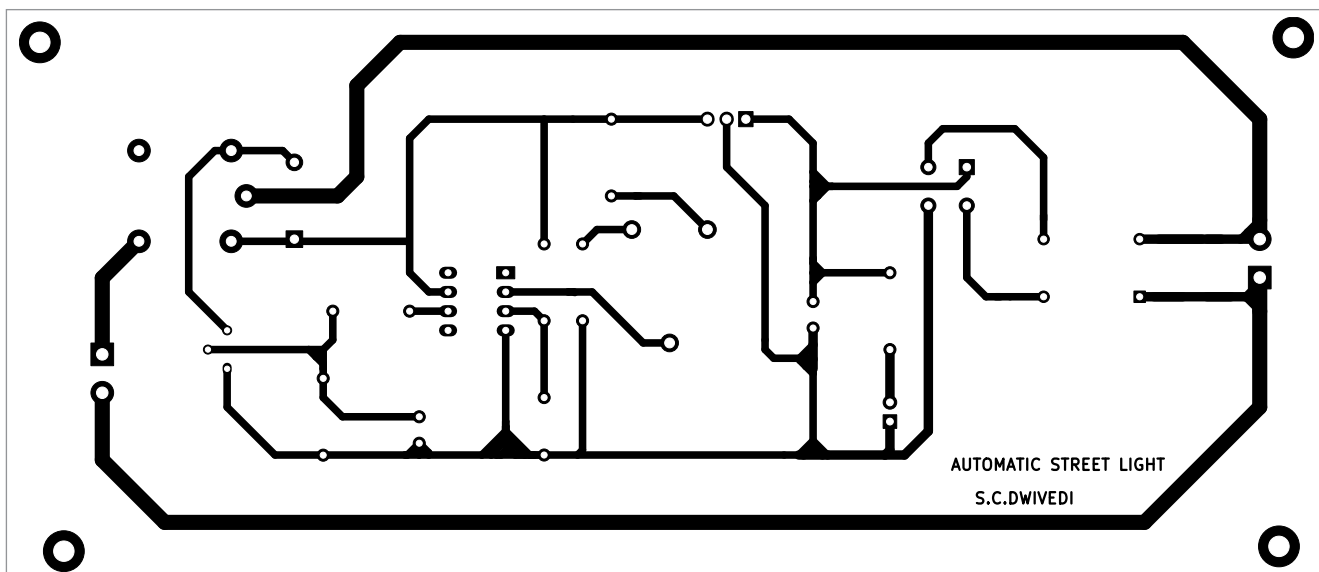


Fig. 3: Actual-size PCB of the circuit

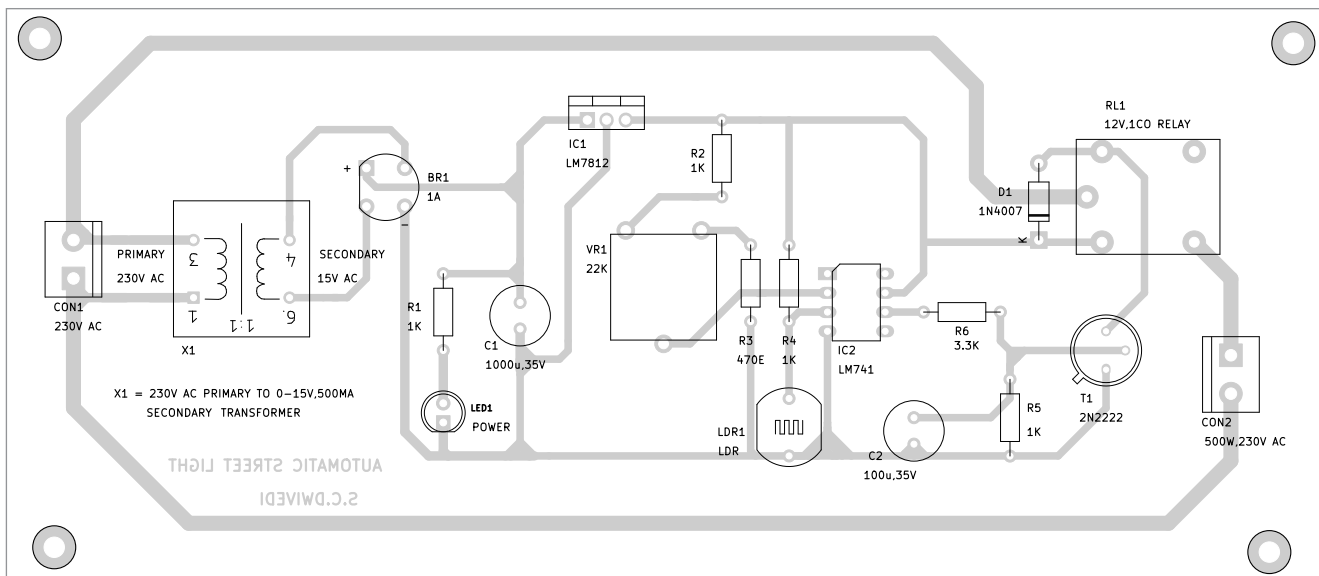


Fig. 4: Component layout of the PCB

put at pin 6 of IC2 is also low. Relay RL1 does not energise in this condition and the streetlight remains off.

As night approaches, light falling on LDR1 reduces. At some point, input at pin 3 of IC2 becomes higher than the reference voltage at pin 2, so output at pin 6 of IC2 goes high. Now relay RL1 energises via relay driver transistor T1. As a result, the streetlight gets 230V AC and switches on.

The streetlight remains on till dawn but in the morning, when sufficient light starts falling on LDR1 once

again, relay RL1 de-energises and switches off the streetlight. Thereafter, the cycle repeats each day.

Construction and testing

An actual-size, single-side PCB layout for the automatic streetlight is shown in Fig. 3 and its component layout in Fig. 4. After assembling the circuit on PCB, enclose it in a suitable box. Fix LED1 in front of the box and connector CON2 on its rear side to connect to the streetlight.

Proper installation of the assem-

bled circuit box is very important. Install the circuit box on the streetlight pillar in such a way that normal daylight falls on LDR1 but light from the streetlight does not fall on it. After proper installation, connect 230V AC mains voltage across CON1. The circuit is now ready to use automatically. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at: <https://www.electronicsforu.com/videos-slideshows/live-diy-automatic-street-light>